An Integrated QKD–PQC Framework for Secure Iris-Based Biometric Authentication

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***Abstract*—As cybersecurity evolves, biometric authentication offers a secure, user-centric approach to identity. However, these systems depend on reusable biometric templates, whose compromise permanently exposes identity. This paper proposes an end-to-end framework for iris biometric authentication, avoiding reusable templates. It integrates iris verification, post-quantum authentication, and quantum-secured session key establishment seamlessly. The proposed architecture employs local iris reconstruction, eliminating the need for reusable biometric templates. It uses ML-DSA for secure terminal-server mutual authentication and signed QAuth session authorisation. T12 biased-base quantum key distribution establishes quantum-secured keys across the communication channel securely. Recursive cascade reconciliation corrects errors, followed by privacy amplification for enhanced key security. HKDF-based derivation generates session keys, while AES-GCM protects subsequent communication between authenticated parties. A FastAPI backend implements the complete end-to-end pipeline for practical prototype evaluation and deployment. The framework is accessed through a React-based graphical interface for interactive system operation. It simulates distance-aware physical-layer conditions, including fibre attenuation and detector inefficiency accurately. The simulation also incorporates intrinsic optical scan noise and dark counts realistically. Performance evaluation includes 6,800 trials using a one-factor-at-a-time experimental study across variable parameters. The investigated factors include communication distance, basis bias, and parameter estimation settings. Additional factors include QBER thresholds, transmission batch sizes, and adversary interception scenarios. The analysis further examines privacy security margins across the proposed authentication framework.**

***Index Terms*—Post quantum cryptography, Quantum Key Distribution, Biometric authentication, Biometric Reconstruction**

I. Introduction

Biometric authentication has become a foundation of modern digital identity systems [1]. Security-sensitive domains have started to replace conventional credentials such as passwords and cards with biometric authentication. Today its use is extended to banking, healthcare, border security, and national digital infrastructure where reliable identification is mandatory [2]. Biometric features such as fingerprints, facial features, and iris patterns are especially useful because they are closely related to an individual. Traditional biometric systems commonly retain digital representations of biometric information that are subsequently required during identity matching. If this stored information gets compromised, it leaks the user identity forever. Fuzzy extractor techniques offer a potential alternative by deriving consistent cryptographic secrets [3] from inherently noisy biometric measurements while accommodating natural variations during acquisition. Protecting the biometric component alone, however, does not secure the complete authentication environment. Beyond biometric protection, communication security must address quantum threats that endanger RSA and ECC, motivating post-quantum cryptography [4]. The Module-Lattice Digital Signature Algorithm (ML-DSA) addresses this requirement by providing quantum-resistant digital authentication, while quantum key distribution (QKD) offers a different capability through the establishment of cryptographic keys based on quantum mechanical principles [5]. These limitations and capabilities make post-quantum cryptography (PQC) and QKD complementary technologies within a broader security architecture.

Existing approaches rarely integrate template-less biometrics, quantum-resistant authentication, and authenticated quantum key establishment end-to-end. Furthermore, practical implementations must account for factors beyond the individual cryptographic components, including optical attenuation, detector inefficiency, reconciliation leakage, and communication latency. Integrating these requirements into one coherent authentication pipeline therefore remains a significant research challenge [6]. To address this gap, this paper presents an integrated post-quantum biometric authentication framework that brings these components together within a unified protocol pipeline. The proposed architecture combines biometric key reconstruction based on fuzzy extraction principles with ML-DSA-based mutual authentication. It additionally incorporates authenticated QAuth session tokens and T12 biased-basis QKD for session establishment [7].

Recursive cascade reconciliation is employed for error correction, followed by privacy amplification and HKDF-based session key derivation. The resulting session keys are subsequently used to protect application-layer communication through AES-GCM presented in Fig. 1. The complete framework is implemented as a functional research prototype with a graphical interface supporting end-to-end execution. Its performance is evaluated under realistic physical-layer conditions involving optical attenuation, detector efficiency,

dark counts, and distance-dependent communication latency. The remainder of this paper details the proposed architecture and its implementation methodology. The subsequent sections discuss biometric reconstruction, post-quantum authentication, quantum key establishment, secure communication, and experimental evaluation.

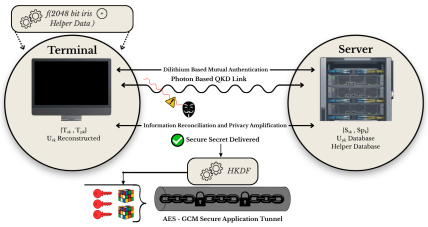


Fig. 1. High Level Diagram for QBioAuth Pipieline

II. Related Work

The following discussion in this section progressively examines the four overlapping directions, of biometric template protection, post quantum authentication, quantum secured communication and hybrid architectures combining them.

A. Biometric Systems

Biometric credentials cannot be replaced because they are inherently associated with an individual. Therefore, maintaining confidentiality during the storage and transmission of biometric templates is essential [8]. Approaches such as cancellable biometrics, fuzzy commitments, and fuzzy extractors have been proposed to address these concerns, with fuzzy extractors deriving reproducible cryptographic material from noisy biometric data using non-secret helper information [9]. The integration of fuzzy commitments with blockchain has further introduced decentralized approaches for biometric protection [10]. However, these approaches primarily address biometric security and do not resolve quantum vulnerabilities affecting the surrounding authentication and communication infrastructure.

B. PQC-Based Authentication

According to NIST standardised ML-DSA approach is used to authenticate the digital signature [11]. Bauspieß *et al.* demonstrated post quantum secure two part computation for iris template protection [12]. However, PQC authenticates computationally, while quantum communication establishes keys with information-theoretic security. It creates the need for complementary architectures where PQC guarantees identity and quantum communication establishes a secure secret key.

C. Quantum Communication

QKD allows two parties to establish a secret key while interception in between is observable through statistical disturbance. BB84 establishes the fundamental prepare-and-measure model for QKD [13] while Lo *et al.* later show improvements in sifting efficiency through asymmetric basis selection [14]. Practical QKD further needs classical post-processing. It includes reconciliation through mechanisms like Cascade [15] and privacy amplification to reduce the adversarial information [16]. However, QKD also requires biometric identity verification and authentication on the classical communication channel.

D. Current Hybrid Systems

Recent works like Qasim *et al.* [17] have begun integrating PQC, quantum communication, and biometrics. They combine ML-KEM, QSDC inspired mutual authentication, and QKD derived keys to encrypt and transmit biometric templates [17]. But their architecture still retains the biometric template as a transmitted object.

III. Proposed System Architecture

The proposed architecture integrates biometric authentication, PQC and QKD into a single end-to-end authentication and secure communication pipeline as discussed below.

Fig. 2 illustrates the proposed authentication framework, comprising user enrollment and authentication phases. Enrollment occurs once, whereas authentication is performed whenever users request secure communication access.unnecessary biometric processing.

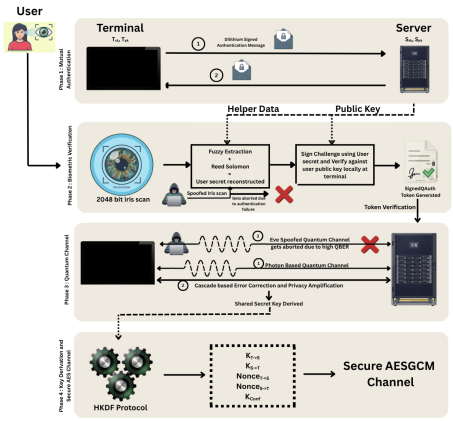


Fig. 2. Overall System Architecture for QBioAuth Pipeline

A. System Initialisation and Terminal-Server Mutual Authentication

During initialization, separate terminal and server entities are created with isolated memory spaces. Each node receives an ML-DSA public-private key pair and communicates only through predefined messages. Every session begins with mutual authentication before any user-specific information is processed.Mutual authentication proceeds as follows:

1) The terminal generates an authentication message containing timestamps and a fresh nonce, signs it using sk_T, and sends

$$\sigma_T = \mathrm{Sign}_{sk_T}(m \parallel t \parallel n). \tag{1}$$

2) The server verifies the signature using pk_T,

$$\text{Verify}_{pk_T}(m, \sigma_T) = 1, \tag{2}$$

and validates the timestamp and nonce for freshness and replay resistance.

3) The server performs the same signed challenge procedure in reverse, which the terminal verifies using the server's public key

Only after both verifications succeed is the endpoints considered mutually authenticated, permitting subsequent biometric enrollment or authentication.

B. User Enrollment

Enrollment process is performed once for each user and creates a long-term cryptographic identity associated with the user's biometric without creating, storing or transmitting a reusable biometric template. For an unregistered user, the terminal performs :

1) The user's unique identifier (email) and iris scan are acquired, and the iris is converted into a stable binary representation for fuzzy construction.
2) The fuzzy construction process derives stable data from which the user's ML-DSA key pair is deterministically generated.
3) The public verification key and non-secret helper data required for future biometric reconstruction are securely transmitted to the authentication server and associated with the user's identity.
4) After successful enrollment, the terminal immediately discards temporary enrollment data, including the iris scan, while the server retains only the public key and helper data.

Since the helper data alone can't reconstruct the enrolled biometric, server side compromise will not expose the user's iris information or private credential.

C. User Authentication

For an enrolled user, authentication reconstructs the biometric-derived signing capability from a fresh iris observation without transmitting the biometric or private-key material. The terminal performs :

1) Using the user's email, the terminal retrieves the associated public verification key and helper data from the authentication server.
2) A fresh iris scan is converted into its binary representation and combined with the helper data through fuzzy reconstruction, tolerating limited acquisition noise.
3) Successful reconstruction locally regenerates the user's ML-DSA private signing key within the trusted terminal.
4) The reconstructed key signs a fresh authentication challenge, whose signature is verified against the retrieved public key.

Successful verification confirms the user's biometric identity without storing or transmitting the biometric template or private key. The protocol then proceeds to QAuth generation for quantum-session authorisation.

D. QAuth Token Generation

Following a successful user authentication, the terminal generates a signed Quantum Authorisation (QAuth) token that conveys cryptographic evidence of authentication to the server before quantum resources are allocated.

1) The terminal constructs

$$QAuth = \{UID, SID, T_{\text{issue}}, T_{\text{exp}}, Status\} \tag{3}$$

containing the authenticated user id, session id, issue and expiration timestamps, and authorisation status.

2) The token is signed using the terminal's private key,

$$\sigma_Q = \text{Sign}_{sk_T}(QAuth), \tag{4}$$

and transmitted to the server.

3) The server verifies the signature, token integrity and timestamp validity before authorising QKD session establishment.

QAuth therefore bridges biometric verification and quantum communication, ensuring that quantum resources are allocated only to authenticated terminals acting for successfully verified users.

E. Quantum Key Distribution for Session Establishment

Following successful QAuth verification, the server initiates T12 biased-basis QKD with the authenticated terminal to establish fresh shared secret material. The protocol proceeds as follows :

1) The terminal and server select their preparation/measurement bases according to

$$P(Z) = p, \qquad P(X) = 1 - p, \tag{5}$$

with $p = 0.8$ in the validated baseline configuration.

2) Quantum states are transmitted through the distance-aware channel model incorporating fibre attenuation, detector inefficiency, intrinsic optical error and dark counts. Fibre transmission follows

$$\eta_{\text{fiber}}(d) = 10^{-\alpha d}. \tag{6}$$

Transmission continues until the required detector events satisfy

$$N_{\text{det}} \geq N_{\text{target}}. \tag{7}$$

3) Both parties exchange signed basis information over the authenticated classical channel and discard mismatched-basis measurements.
4) A fixed fraction of the sifted key is disclosed for parameter and QBER estimation, with

$$QBER = \frac{N_{\text{error}}}{N_{\text{compared}}}. \tag{8}$$

Sessions exceeding the configured QBER threshold are aborted and a fresh quantum transmission is attempted.

The remaining sifted key is shared but may contain residual disagreements caused by channel and detector imperfections. Therefore the sifted bits proceed to information reconciliation before cryptographic use.

F. Cascade-Based Information Reconciliation

The sifted keys may contain residual disagreements from channel and detector noise and therefore cannot yet be used as shared cryptographic material. The framework applies Cascade-based reconciliation over the authenticated classical channel as follows:

1) The sifted key is partitioned into blocks B_i, and each party computes
$$P(B_i) = \bigoplus_{j=1}^{n_i} b_j. \tag{9}$$
2) For blocks satisfying
$$P_T(B_i) \neq P_S(B_i), \tag{10}$$
binary search recursively divides the block to identify and correct the erroneous bit.
3) Multiple reconciliation passes with recursive look-back correct residual disagreements missed in earlier passes until
$$K_T = K_S. \tag{11}$$

Reconciliation produces identical terminal and server keys but publicly reveals parity information. The reconciled secret therefore undergoes privacy amplification before cryptographic use.

G. Privacy Amplification

To reduce information potentially exposed through parameter estimation, reconciliation, or quantum-channel observation, the reconciled secret is compressed through privacy amplification:

1) The reconciled secret is hashed as
$$K_{PA} = \mathrm{SHA256}(K_{REC}). \tag{12}$$
2) The retained secret length is determined by
$$L_{PA} = L_{REC} - \lambda, \tag{13}$$
where λ is the configured security margin accounting for information leakage.
3) Both endpoints independently obtain the identical privacy-amplified secret,
$$K_{PA} = K_{PA}^{(T)} = K_{PA}^{(S)}. \tag{14}$$

The resulting shared secret is subsequently passed to HKDF to derive independent cryptographic material for application-layer communication.

H. HKDF-Based Key Derivation and Classical Channel Communication

The privacy-amplified secret is expanded through HKDF into independent, purpose-specific session material:

$$\{K_{TS}, K_{ST}, N_{TS}, N_{ST}, K_{CONF}\} = \mathrm{HKDF}(K_{PA}, Salt, Info) \tag{15}$$

where K_{TS} and K_{ST} are direction-specific encryption keys, N_{TS} and N_{ST} are the corresponding nonce bases, and K_{CONF} confirms that both endpoints derived identical key material.

Application-layer communication then proceeds as follows:

1) Terminal-to-server and server-to-terminal messages use their independently derived keys and nonce material.
2) AES-GCM provides authenticated encryption for each message,
$$(C, \tau) = \text{AES-GCM}_{K_{enc}}(M), \tag{16}$$
producing ciphertext C and authentication tag τ from plaintext M.

The resulting channel provides confidential and authenticated application-layer communication using session material derived from the reconciled and privacy-amplified quantum secret.

I. System Assumptions and Limitations

The framework assumes trusted enrollment and terminal execution, while biometrics, quantum transmission and detector behaviour are simulated rather than hardware-derived. Hardware side channels, physical terminal compromise, malicious insiders and denial-of-service attacks remain outside the threat model. Consequently, the reported distance and reliability results characterize the simulated configuration and should not be interpreted as hardware performance limits.

IV. Experimental Methodology

The proposed framework was evaluated step by step in a controlled simulation experiment environment to identify the effects and trustworthiness of the framework when physical conditions, quantum channel conditions and noises are varied.

A. Experimental Setup

According to the implementation details mentioned in the Section III, a complete experiment study was conducted. Unless varied as a part of the simulation testing or experiment factor, the protocol and physical channel parameters are described in Table I. This creates a common operation ground from which the individual effects of selected parameters can be measured.

B. One-Factor-at-a-Time Experimental Study

A One-Factor-at-a-Time (OFAT) study varied one parameter while holding all others at the baseline in Table I, enabling isolation of its effects on security, reliability and resource efficiency. Seven factors spanning 27 experimental cases were evaluated as summarised in Table II, covering physical-channel conditions, QKD configuration, adversarial interception and post-processing. A fixed master seed ensured reproducibility. Each case was evaluated over 100 trials without Eve and 100 trials under 100% interception, except the Eve-interception-fraction sweep itself, resulting in 6,800 individual experimental records.

TABLE I
BASELINE CONFIGURATION OF THE PROPOSED FRAMEWORK

Parameter	Assumption / Role	Baseline
Z-basis probability	Probability of selecting the dominant T12 basis	80%
Parameter-estimation fraction	Fraction of sifted bits disclosed for QBER estimation	15%
QBER abort threshold	Maximum estimated QBER permitted before session rejection	8%
Target detector events	Minimum accumulated detections before QKD processing	8192
Transmission batch size	Optical pulses represented in each acquisition batch	8192
Source emission efficiency	Probability of successful photon emission	100%
Coupling efficiency	Efficiency of coupling photons into the optical channel	80%
Detector efficiency	Mean photon detection efficiency	20%
Intrinsic optical error	Mean baseline optical error probability	1%
Dark-count probability	Probability of a spurious detector event per gate	10^{-6}
Dark-count error probability	Probability that a dark-count detection produces an incorrect bit	50%
Cascade passes	Configured recursive reconciliation passes	6
Privacy security margin	Additional key-length deduction during privacy amplification	128 bits
Maximum QKD attempts	Maximum fresh QKD attempts following recoverable failure	3
Synthetic iris noise	Bit-error probability applied to a repeated synthetic iris scan	1%
Synthetic iris rotation	Circular displacement used to model acquisition misalignment	-4 bits
QAuth validity	Maximum authorisation-token lifetime	120 s

TABLE II
PARAMETERS EVALUATED IN THE OFAT STUDY

Factor	Evaluated Values
Communication distance	0, 20, 50, 80, 100, 150, 200 km
Transmission batch size	1024, 2048, 4096, 8192, 16384 pulses
Z-basis probability	50%, 70%, 80%, 85%, 93.75%
Parameter-estimation fraction	3%, 5%, 10%, 15%, 20%
QBER abort threshold	3%, 5%, 8%, 10%, 12%
Eve interception fraction	0%, 10%, 25%, 50%, 75%, 100%
Privacy security margin	64, 128, 192, 256 bits

C. Evaluation Metrics

Evaluation records session success, QBER, transmitted pulses and acquisition rounds, sifted/reconciled key lengths, corrected errors, reconciliation leakage, privacy-amplified secret length and end-to-end latency. Protocol reliability is measured as

$$S_{\text{rate}} = \frac{N_{\text{successful}}}{N_{\text{total}}} \times 100 \tag{17}$$

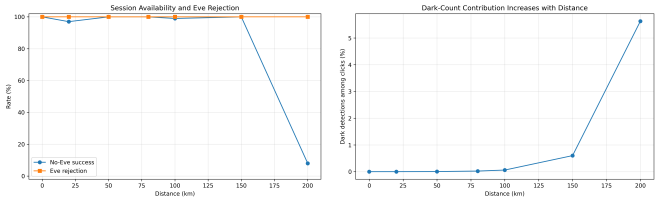
while post-processing efficiency is characterised by the final secret-retention ratio

$$R_{\text{secret}} = \frac{L_{PA}}{L_{\text{sifted}}}. \tag{18}$$

Optical acquisition latency is additionally recorded separately from end-to-end latency to distinguish quantum-channel degradation from classical authentication, reconciliation and key-derivation overheads.

V. RESULTS AND DISCUSSION

The OFAT study conducted shows that the proposed framework runs stably around the baseline operating region. The behaviour of the entire protocol also changes predictably as the baseline metrics change when the physical channel loss, fraction of Eve intercepted channel and protocol specific parameters are increase.



(a) (b)

Fig. 3. Impact of Distance on QKD performance: (a) Effect of Distance on Session Success and Acceptance (b) effect of Distance on Dark Counts.

A. Effect of Communication Distance

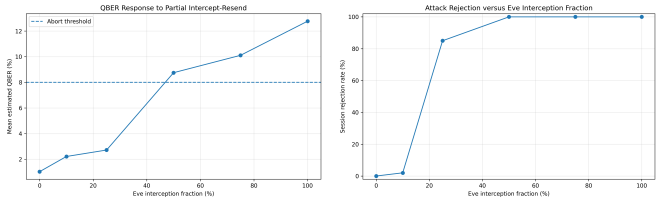
The physical layer was most strongly affected by the increase in communication distance. Without an Eve attack, the session success remained at 97-100% of channel success between 0-150 km (Figure 3a). But under the same conditions, the channel success fell to 8% at 200 km distance while the mean transmitted pulse for channel completion increased from 3.09×10^4 at 0 km to 2.54×10^7 at 150 km and 2.43×10^8 at 200 km. The fibre attenuation exponentially increased the transmission requirements rather than the QBER. The mean QBER remains approximately 1% through 100 km before increasing to 1.29% at 150 km and 3.96% at 200 km as dark counts become non negligible 3b. The mean privacy amplified secret bitstring length remains 1.28-1.36 kbits until 100 km, decreases slightly to 1.17 kbits at 150 km and collapses due to low session success at 200 km. These results together show that 150 km is a practical upper operating region for the simulated baseline configuration.

B. Effect of Basis Bias in the QKD

According to the T12 protocol and the results of the analysis, increasing the Z-basis probability improves sifting efficiency and the basis-match rate increases from 50.1% at $P(Z) = 0.5$ to 68.1% at the $P(Z) = 0.8$ baseline and 88.2% at $P(Z) = 0.9375$, with mean sifted-key lengths increasing from 2051 to 2788 and 3614 bits, respectively. However, over increasing the bias reduces adversary detection. Under 100% interception, all sessions are rejected for $P(Z) \leq 0.85$, whereas approximately 5% succeeded at $P(Z) = 0.9375$. The mean estimated QBER simultaneously decreases from 25.6% under unbiased selection to 2.33% at the strongest bias. Thus, $P(Z) = 0.8$ provides a stable trade off between T12 sifting efficiency and reliable interception detection.

C. Adversarial Interception Behaviour

Increasing intercept-resend activity produces a clear security response. Mean QBER increases from 1.03% without Eve to 2.21% at 10% interception and continued rising with interception fraction (Fig. 4a). Consequently, session success decreased from 100% without Eve to 98% at 10%, 15% at 25%, and 0% for interception fractions of 50% or higher (Fig. 4b). Surviving secret material decreased simultaneously, with mean privacy-amplified key length falling from approximately 1312 bits without Eve to 729 bits at 10% interception



(a) (b)

Fig. 4. Impact of Eve on QKD performance: (a) Effect of Eve on QBER and (b) effect of Eve on channel success.

and approximately 49 bits at 25%. Thus, interception is reflected not only in QBER-based rejection but also in sharply reduced session reliability and usable secret material.

The remaining OFAT parameters primarily affected resource efficiency and post-processing trade-offs. Increasing batch size reduced acquisition rounds without significantly changing total transmitted pulses, indicating that channel loss dominates optical cost. Increasing parameter estimation improved QBER sampling but reduced retained key material, while the QBER threshold balanced legitimate-session acceptance and adversarial rejection. A larger privacy margin reduced final secret length approximately linearly without materially affecting session success, supporting the baseline of 15% parameter estimation, an 8% QBER threshold, and a 128-bit privacy margin. Recursive Cascade reconciliation was highly reliable but showed residual failures across 6,800 trials, concentrated in the distance, basis bias, and QBER-threshold groups, motivating explicit post-reconciliation key confirmation. Overall, QBioAuth maintained reliable authenticated sessions under the validated baseline, with distance as the dominant physical constraint and other parameters defining security-efficiency trade-offs. Thus, the selected baseline represents a balanced operating point rather than a universal optimum.

VI. Conclusion and Future Work

This work presents an end-to-end quantum-secured biometric authentication and communication framework integrating local biometric reconstruction and ML-DSA-based device and user authentication. Further QAuth session authorisation, biased-basis QKD, recursive cascade reconciliation, privacy amplification, HKDF key derivation and AES-GCM protected communication without transmitting reusable biometric templates. Experimental evaluation demonstrated high legitimate-session reliability up to approximately 150 km under the validated baseline configuration, while revealing measurable security-efficiency trade-offs. In particular, aggressive basis bias increased sifting efficiency but weakened disturbance observability, whereas increasing adversarial interception progressively reduced session success and usable secret material.

The present implementation remains an experimental prototype and can be enhanced as a future direction by replacing synthetic iris codes with real sensor data. It can be evaluated by fuzzy reconstruction under realistic biometric variations while strengthening the trusted terminal through hardware-backed execution, secure memory handling and side-channel resistance. Quantum communication can similarly be validated using experimental optical hardware and detectors. Further extensions include scalable QKD networking through trusted or quantum-repeater-supported architectures, distributed key management, and continuous biometric authentication for periodically revalidating user presence during established sessions.

References

[1] S. Kakade and U. Raut, "Biometric authentication systems: Trends, challenges, and future prospects – a comprehensive review," *MethodsX*, vol. 16, p. 103908, 2026.

[2] D. Temoshok, Y.-Y. Choong, A. Regenscheid, R. Galluzzo, J. L. Fenton, J. Richer, and N. Lefkovitz, "Digital identity guidelines: Authentication and authenticator management," National Institute of Standards and Technology, NIST Special Publication 800-63B-4, Jul. 2025.

[3] Y. Dodis, L. Reyzin, and A. Smith, "Fuzzy extractors: How to generate strong keys from biometrics and other noisy data," in *SIAM Journal on Computing*, vol. 38, no. 1, 2008, pp. 97–139.

[4] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM Journal on Computing*, vol. 26, no. 5, pp. 1484–1509, 1997.

[5] N. Gisin, G. Ribordy, W. Tittel, and H. Zbinden, "Quantum cryptography," *Reviews of Modern Physics*, vol. 74, no. 1, pp. 145–195, 2002.

[6] National Institute of Standards and Technology, "Module-lattice-based digital signature standard," National Institute of Standards and Technology, Federal Information Processing Standard 204, 2024.

[7] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," in *Proceedings of IEEE International Conference on Computers, Systems and Signal Processing*, Bangalore, India, 1984, pp. 175–179.

[8] *ISO/IEC 24745:2022 Information Security, Cybersecurity and Privacy Protection—Biometric Information Protection*, International Organization for Standardization Std., 2022.

[9] Y. Dodis, R. Ostrovsky, L. Reyzin, and A. Smith, "Fuzzy extractors: How to generate strong keys from biometrics and other noisy data," *SIAM Journal on Computing*, vol. 38, no. 1, pp. 97–139, 2008.

[10] N. A. Alzahab, G. Rafaiani, M. Battaglioni, F. Chiaraluce, and M. Baldi, "Decentralized biometric authentication based on fuzzy commitments and blockchain," in *Proceedings of the 6th International Conference on Blockchain Computing and Applications (BCCA)*, 2024, pp. 64–72.

[11] National Institute of Standards and Technology, "Module-lattice-based digital signature standard," National Institute of Standards and Technology, Tech. Rep. FIPS 204, 2024.

[12] P. Bauspieß, J. Kolberg, D. Demmler, J. Krämer, and C. Busch, "Post-quantum secure two-party computation for iris biometric template protection," in *2020 IEEE International Workshop on Information Forensics and Security (WIFS)*, 2020, pp. 1–6.

[13] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing," *Theoretical Computer Science*, vol. 560, pp. 7–11, 2014, originally presented at the IEEE International Conference on Computers, Systems and Signal Processing, Bangalore, 1984.

[14] H.-K. Lo, H. F. Chau, and M. Ardehali, "Efficient quantum key distribution scheme and a proof of its unconditional security," *Journal of Cryptology*, vol. 18, no. 2, pp. 133–165, 2005.

[15] G. Brassard and L. Salvail, "Secret-key reconciliation by public discussion," in *Advances in Cryptology—EUROCRYPT '93*, ser. Lecture Notes in Computer Science, vol. 765. Springer, 1994, pp. 410–423.

[16] C. H. Bennett, G. Brassard, C. Crépeau, and U. M. Maurer, "Generalized privacy amplification," *IEEE Transactions on Information Theory*, vol. 41, no. 6, pp. 1915–1923, 1995.

[17] T. Qasim, V. A. Siris, I. Oosthuizen, M. Rajarajan, and S. Biswas, "Quantum secure biometric authentication in decentralised systems," in *2025 IEEE International Joint Conference on Biometrics (IJCB)*. IEEE, 2025.